

**1-2: Atomic & Molecular Structure (14–18%)****Electron Configuration & Periodic Trends**

**Aufbau principle:** Fill orbitals in order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p ... **Hund's rule:** Electrons pair-up only after single electrons occupy each orbital (maximize unpaired electrons). **Pauli exclusion:** max 2 electrons per orbital (opposite spins).

**Periodic trends** (left → right, top → bottom): Atomic radius ↑ down, ↓ right. Ionization energy ↑ right, ↑ up. Electron affinity ↑ right (trends broken for N, O). Electronegativity ↑ right, ↑ up.

**Exceptions:** Cr, Mo, Cu: d<sup>10</sup>s<sup>1</sup> (stability).

Lanthanide/actinide f-block: fill after 6s. **Ion electrons:** Add electrons for anions, subtract for cations. **Shielding:** Inner electrons ↑ → outer electrons ↑ binding energy ↓, atomic radius ↑.

**Lewis Structures & Formal Charge**

**Steps:** count valence e<sup>-</sup> (group number for main group). Draw skeleton (least H in middle). Add e<sup>-</sup> to each atom (octet rule). If short, form double/triple bonds. **Formal charge:** FC = V – N – B/2 (V = valence, N = nonbonding, B = bonding e<sup>-</sup>).

**Resonance:** Multiple valid Lewis structures differing only in e<sup>-</sup> position, NOT atoms. Real structure is weighted average. Major resonance: minimizes FC magnitude & puts negative charge on more electronegative atom. Dashed bonds indicate delocalization.

**VSEPR Geometry & Hybridization**

**Hybridization:** Linear = sp. Trigonal = sp<sup>2</sup>. Tetrahedral = sp<sup>3</sup>. Trig bipyramidal = sp<sup>3</sup>d. Octahedral = sp<sup>3</sup>d<sup>2</sup>. **Lone pairs occupy hybrid orbitals first.** Angle compression: lone pairs > bonding pairs in repulsion.

✓ Lewis & VSEPR separate: Lewis shows e<sup>-</sup>, VSEPR predicts shape from electron pair geometry. Multiple resonance? Draw most stable; real is hybrid. Electronegativity difference > 1.7 → ionic; 0.4–1.7 → polar covalent; < 0.4 → nonpolar.

**Thermodynamics (Unit 6, 7–9%)****First Law & Heat Calculations**

**First Law:** ΔU = q + w. Internal energy change = heat absorbed + work done ON system. **Calorimetry:** q = mcΔT (liquid, solid; c = specific heat, J/g·°C). **ΔT:** Final – Initial. Heat capacity C = m·c → q = C·ΔT.

**Bomb calorimeter (const volume):** w = 0 → ΔU = q<sub>bomb</sub> = (–q<sub>solution</sub>). Measure ΔT of water; back-calc ΔU<sub>rxn</sub> = –q/n<sub>rxn</sub> (J/mol). **Coffee cup calorimeter (const P, open):** w ≈ 0 → ΔH ≈ q.

**Mixing/neutralization:** q<sub>lost</sub> + q<sub>gained</sub> = 0. m<sub>1</sub>c<sub>1</sub>ΔT<sub>1</sub> + m<sub>2</sub>c<sub>2</sub>ΔT<sub>2</sub> = 0 (one ↑ temp, one ↓ temp, sum = 0). No heat exchange with surroundings (isolated system).

**Enthalpy & Hess's Law**

**Enthalpy:** H = U + PV. At const P: ΔH = q<sub>p</sub> (heat at constant pressure). **ΔH<sub>rxn</sub> = Σ ΔH<sub>f</sub>(products) – Σ ΔH<sub>f</sub>(reactants).** ΔH<sub>f</sub> = enthalpy of formation (kJ/mol). Element in natural state: ΔH<sub>f</sub> = 0.

**Hess's Law:** ΔH<sub>rxn</sub> is sum of ΔH for intermediate steps, regardless of path. Add/subtract given equations to get target. **Reverse equation → negate ΔH. Multiply by n → multiply ΔH by n.** Example: (1) A → B, ΔH<sub>1</sub> = 100. (2) B → C, ΔH<sub>2</sub> = 50. Net: A → C, ΔH = 100 + 50 = 150 kJ.

**Bond Energy & Reaction Energy**

**Bond energy (BE):** Energy required to break bonds (endothermic, +). ΔH<sub>rxn</sub> = BE<sub>broken</sub> – BE<sub>formed</sub>. **Exothermic:** strong bonds formed, weak bonds broken (ΔH < 0). **Endothermic:** weak bonds formed, strong bonds broken (ΔH > 0).

**Example:** H<sub>2</sub> + F<sub>2</sub> → 2HF. H–H (436 kJ/mol) + F–F (158) broken. 2 H–F (567 kJ/mol) formed. ΔH = (436+158) – 2(567) = 594 – 1134 = –540 kJ. Very exothermic.

✓ FRQ enthalpy: (1) Use ΔH<sub>f</sub> if given; (2) Use Hess's law if multiple steps; (3) Show signs (exo–, endo+); (4) State work/heat flow direction (system gains/loses heat). Always include units & magnitude.

**3: Intermolecular Forces & Properties (18–22%)**

✓ IMF decision tree: H bonded to F/O/N? → H-bond. Charged? → Ion-dipole. Both polar? → Dipole-dipole. Both nonpolar? → London dispersion. Always list strongest first.

**IMF Types & Physical Properties**

**IMF ranking (strongest → weakest):** Ion-dipole > H-bond (F/O/N) > Dipole-dipole > London dispersion. **H-bonds:** H bonded to F/O/N. DNA, ice, water. Formation requires: electronegativity diff + lone pair on F/O/N. **LDF:** Present in ALL molecules; ↑ with surface area, molar mass & e<sup>-</sup> count. Temporary dipoles from electron fluctuations.

**Physical property trends:** ↑ IMF → ↑ bp, ↑ mp, ↓ Pvap, ↑ viscosity, ↑ surface tension. Water anomaly: H-bonds make ice less dense (floats). Solubility: polar solutes in polar solvents; nonpolar in nonpolar.

**Colligative Properties**

**Four colligative properties depend on particle count (not identity):** (1) **Freezing point depression:** ΔT<sub>f</sub> = K<sub>f</sub>m-i. Solvent freezes at lower T. (2) **Boiling point elevation:** ΔT<sub>b</sub> = K<sub>b</sub>m-i. Solvent boils at higher T. (3) **Osmotic pressure:** π = MRT (uses molarity, not molality!). Reflects tendency of solvent to enter solution across semipermeable membrane. (4) **Vapor pressure lowering:** ΔP = χ<sub>solute</sub>·P<sub>°solute</sub>. Raoult's law for ideal solutions. **Molality m = mol solute / kg solvent** (NOT kg solution). Temperature-independent, used for colligative calcs.

**Electrolyte factor i:** Accounts for dissociation. NaCl i ≈ 2, CaCl<sub>2</sub> i ≈ 3, weak acid/base i ∈ 1–2 (partial dissociation), nonelectrolyte i = 1. Real i < ideal due to ion pairing & activity coefficients.

**MW from colligative:** Freezing point: m = ΔT<sub>f</sub>/K<sub>f</sub>, then MW = (grams)/(m × kg solvent). Osmotic pressure: MW = (grams × RT)/(π × V). K<sub>f</sub> = 1.86 °C/m, K<sub>b</sub> = 0.512 °C/m, R = 0.0821 L·atm/(mol·K).

Δ Colligative uses MOLALITY (mol/kg solvent), NOT molarity. Pure solids/liquids omitted from K expressions.

**Equilibrium (Unit 7, 7–9%)****K Expression & ICE Table**

**Equilibrium constant:** For reaction aA + bB ⇌ cC + dD: K<sub>c</sub> = [C]<sup>c</sup>[D]<sup>d</sup> / [A]<sup>a</sup>[B]<sup>b</sup>. **Interpretation:** K > 1: products favored at eq. K < 1: reactants favored. K<sub>reverse</sub> = 1/K<sub>forward</sub>. K<sub>p</sub> uses partial pressures (atm): K<sub>p</sub> = K<sub>c</sub>(RT)<sup>Δn</sup>, where Δn = (moles gas products) – (moles gas reactants), R = 0.0821, T = Kelvin.

**ICE (Initial, Change, Equilibrium) table:** Row 1: Initial conc (given or 0 if not present). Row 2: Change due to rxn (–x for reactants consumed, +x for products formed). Row 3: Equilibrium = I + C. Substitute E row into K expression, solve for x (use quadratic formula or assume x negligible if K >> 1 or K << 1, check validity with 5% rule).

**Q vs K & Solubility**

**Reaction quotient Q:** Same expression as K but uses current/arbitrary concentrations (NOT necessarily at equilibrium). **Directional prediction:** If Q < K: reaction shifts RIGHT (forward), consuming reactants & producing products until equilibrium achieved. If Q > K: shifts LEFT (reverse), converting products back to reactants. If Q = K: system at equilibrium, no shift. Q serves as diagnostic tool for predicting rxn direction.

**Solubility equilibria & K<sub>sp</sub>:** For dissolution, e.g., AgCl(s) ⇌ Ag<sup>+</sup>(aq) + Cl<sup>–</sup>(aq), K<sub>sp</sub> = [Ag<sup>+</sup>][Cl<sup>–</sup>]. Solids/pure liquids absent from K<sub>sp</sub>. **Precipitation prediction:** Calculate ion product Q<sub>ion</sub> = [Ag<sup>+</sup>][Cl<sup>–</sup>] from given concentrations. If Q<sub>ion</sub> > K<sub>sp</sub>: precipitate forms (rxn shifts left to decrease ion conc). If Q<sub>ion</sub> < K<sub>sp</sub>: solution unsaturated, dissolution occurs. **Common ion effect:** Adding Cl<sup>–</sup> increases Q<sub>ion</sub>, shifts equilibrium LEFT, reducing [Ag<sup>+</sup>] (lowers AgCl solubility).

✓ FRQ: (1) Write K (omit solids/liquids). (2) Set up ICE with variable x. (3) Solve K = (I+C)/(I–C) for x (approximate valid if 5% rule holds). (4) Use Q vs K to predict shift. (5) K<sub>sp</sub>: compare Q<sub>ion</sub> to predict precipitation or dissolution.

**Chemical Reactions (Unit 4, 7–9%)**

Δ Spectator ions cancel in net ionic. Write full ionic, cross out same ions both sides. Net ionic = reacting species only.

**Stoichiometry & Limiting Reagent**

**Stoichiometry workflow:** (1) Balance equation. (2) Convert grams to moles: n = grams/MW. (3) Use stoichiometric coefficient ratio to find moles product. (4) Convert moles to grams: grams = moles × MW. **Limiting reactant:** Calculate moles product from EACH reactant independently; the smallest value determines the limit. % **yield:** (actual yield / theoretical yield) × 100%.

**Oxidation States & Redox Balancing**

**OS assignment rules:** Element (uncombined) = 0. Monatomic ion = charge of ion. Oxygen = –2 (exception: peroxides H<sub>2</sub>O<sub>2</sub> = –1). Hydrogen = +1 (exception: hydrides e.g., NaH = –1). Fluorine = –1 always. Sum of OS in neutral compound = 0; in polyatomic ion = charge.

**Redox:** OS ↑ = oxidation (lose e<sup>-</sup>). OS ↓ = reduction (gain e<sup>-</sup>). **Half-reaction method:** Split oxidation/reduction. Balance atoms except O/H. Add H<sub>2</sub>O for O, H<sup>+</sup> for H. Balance charge with e<sup>-</sup>. Multiply halves to cancel e<sup>-</sup>, combine.

**Net Ionic Equations & Solubility**

**Net ionic 3-step:** (1) Molecular: all as formulas. (2) Ionic: split aq ions only; keep solids/liquids whole. (3) Cancel spectators. Example: HCl + NaOH → H<sup>+</sup> + OH<sup>–</sup> → H<sub>2</sub>O (net). **Solubility rules:** *Always soluble:* Group 1, NH<sub>4</sub><sup>+</sup>, NO<sub>3</sub><sup>–</sup>, ClO<sub>4</sub><sup>–</sup>. *Insoluble:* Most OH<sup>–</sup>, CO<sub>3</sub><sup>2–</sup>, PO<sub>4</sub><sup>3–</sup>, S<sup>2–</sup> (except Group 1). SO<sub>4</sub><sup>2–</sup> insoluble with Ba<sup>2+</sup>, Pb<sup>2+</sup>. **Combustion:** C<sub>x</sub>H<sub>y</sub> + O<sub>2</sub> → CO<sub>2</sub> + H<sub>2</sub>O; incomplete → CO.

✓ FRQ: (1) Balance molecular equation. (2) Write full ionic (split aqueous salts). (3) Identify & cancel spectators. (4) Write net ionic with state labels. For redox: show half-rxns, balance atoms/charge/e<sup>-</sup>, combine correctly.

**Acids & Bases (Unit 8, 11–15%)**

✓ Henderson-Hasselbalch: pH = pK<sub>a</sub> + log([A<sup>–</sup>]/[HA]) (buffer region only). Strong: [H<sup>+</sup>] = M. K<sub>a</sub> × K<sub>b</sub> = 10<sup>–14</sup>. Titration equiv pH depends on salt (7 for strong-strong, > 7 for weak acid + strong base).

**Strong vs Weak & K<sub>a</sub>/K<sub>b</sub>**

**Strong acids (100% ionization):** HCl, HBr, HI, HNO<sub>3</sub>, H<sub>2</sub>SO<sub>4</sub>, HClO<sub>4</sub>. **Weak acids:** All others, use K<sub>a</sub> + ICE. **Strong bases:** Group 1 OH, Ba(OH)<sub>2</sub>. **Weak bases:** NH<sub>3</sub>, amines, use K<sub>b</sub> + ICE.

**K<sub>w</sub> = [H<sup>+</sup>][OH<sup>–</sup>] = 10<sup>–14</sup>. pH = –log[H<sup>+</sup>]. pH + pOH = 14. K<sub>a</sub> × K<sub>b</sub> = K<sub>w</sub>. Salt hydrolysis: strong acid + weak base → acidic; weak acid + strong base → basic.**

**pH Calculations & Buffers**

**Strong acid/base:** 100% ionization. [H<sup>+</sup>] = M<sub>acid</sub> (ignore water's H<sup>+</sup>). pH = –log[H<sup>+</sup>].

**Weak acid HA:** Set ICE: HA ⇌ H<sup>+</sup> + A<sup>–</sup>. K<sub>a</sub> = [H<sup>+</sup>][A<sup>–</sup>]/[HA]. Assume x << C if K<sub>a</sub> << 0.05C, then [H<sup>+</sup>] = √(K<sub>a</sub>·C). Weak base B: B + H<sub>2</sub>O ⇌ BH<sup>+</sup> + OH<sup>–</sup>. [OH<sup>–</sup>] = √(K<sub>b</sub>·C), pOH = –log[OH<sup>–</sup>], pH = 14 – pOH.

**Buffer:** Weak acid + conjugate base (or weak base + conjugate acid). **Henderson-Hasselbalch:** pH = pK<sub>a</sub> + log([A<sup>–</sup>]/[HA]). Resists pH change when ≈ pK<sub>a</sub> ± 1. Buffer capacity highest near pK<sub>a</sub>. Adding small acid/base shifts ratio slightly.

**Titration Curves & Equivalence Points**

**Equivalence point:** Moles acid = moles base added. **Strong + strong:** pH > 7 (salt neutral). **Weak acid + strong base:** pH > 7 (conjugate base A<sup>–</sup> weak base, hydrolyzes). **Weak base + strong acid:** pH < 7 (conjugate acid BH<sup>+</sup> weak acid, acidic). **Half-equivalence:** pH = pK<sub>a</sub> exactly (since [A<sup>–</sup>] = [HA], log ratio = 0).

Δ Weak acid approximation valid if x/C<sub>o</sub> < 5%. Check: K<sub>a</sub> × 100 / C<sub>o</sub>, if < 5, use approx. Titration shape: flat start, steep at equiv, flat end.

**Kinetics (Unit 5, 7–9%)**

✓ Rate law from EXPERIMENT, NEVER coefficients. Order = sum of exponents. Integrated: 0th = linear, 1st = ln, 2nd = 1/[]. t<sub>1/2</sub> = 0.693/k (1st order only).

**Rate Laws & Order Determination**

rate = k[A]<sup>x</sup>[B]<sup>y</sup>. x, y = order (NOT stoich). Order: 0th (const), 1st ([A]<sup>1</sup>), 2nd ([A]<sup>2</sup>). Overall order = x + y. k units: M<sup>x</sup>(1–order)·s<sup>–1</sup>.

Experimental determination from data: Compare trials where [A] doubles: rate doubles → 1st order, rate x4 → 2nd order, rate unchanged → 0th order.

| Order | Integ. Law                      | Linear Plot | t <sub>1/2</sub>       | k units                         |
|-------|---------------------------------|-------------|------------------------|---------------------------------|
| 0th   | [A] = [A] <sub>o</sub> – kt     | [A] vs t    | [A] <sub>o</sub> /2k   | M/s                             |
| 1st   | ln[A] = ln[A] <sub>o</sub> – kt | ln[A] vs t  | 0.693/k                | s <sup>–1</sup>                 |
| 2nd   | 1/[A] = 1/[A] <sub>o</sub> + kt | 1/[A] vs t  | 1/(k[A] <sub>o</sub> ) | M <sup>–1</sup> s <sup>–1</sup> |

**Activation Energy & Mechanisms**

**Arrhenius:** k = A·e<sup>–(E<sub>a</sub>/RT)</sup>. ln k = ln A – (E<sub>a</sub>/R)·(1/T). Plot ln(k) vs 1/T → slope = –E<sub>a</sub>/R. Higher T → faster k. Catalyst: Lowers E<sub>a</sub>, ↑k without changing ΔH.

RDS (rate-determining step): Slowest elementary step controls overall rate. Rate law matches RDS molecularity. Exponents from RDS stoichiometry, NOT overall coefficients.

**Catalyst vs intermediate:** Catalyst consumed then regenerated (appears in mechanism, NOT net rxn). Intermediate produced then consumed (never appears in overall equation).

Δ Rate law exponents from RDS, NOT balanced coefficients. Intermediates never in overall equation. Mechanism must sum to overall, intermediates cancel.

✓ FRQ: (1) Order from data (doubling method or integrated forms). (2) Rate law & k units. (3) Mechanism: identify RDS, check steps sum correctly.

**Applications of Thermodynamics (Unit 9, 7–9%)****Gibbs Free Energy & Spontaneity**

**Gibbs equation:** ΔG = ΔH – TΔS (T in Kelvin). ΔG < 0: spontaneous forward. ΔG > 0: non-spontaneous (reverse is spont). ΔG = 0: at equilibrium. **Relationship to K:** ΔG° = –RT ln K (R = 8.314 J/mol·K). If K > 1 → ΔG° < 0 → spont. If K < 1 → ΔG° > 0 → non-spont.

**Electrochemistry: Redox & Cell Potentials**

**Galvanic (voltaic) cells (spontaneous):** Anode = oxidation site (–), Cathode = reduction site (+). Standard cell potential: E°<sub>cell</sub> = E°<sub>cathode</sub> – E°<sub>anode</sub> (reduction potentials from tables). **Cell notation:** Anode compartment | salt bridge || Cathode compartment. **Prediction:** E°<sub>cell</sub> > 0 → spontaneous. E°<sub>cell</sub> < 0 → non-spontaneous (electrolytic).

**Nernst equation (25°C):** E<sub>cell</sub> = E°<sub>cell</sub> – (0.0592/n) log Q. n = moles e<sup>–</sup> transferred. Q = [products]/[reactants] (Nernst Q uses conc/pressure). At equilibrium: E = 0, Q = K. **Fundamental link:** ΔG° = –nFE°<sub>cell</sub>, where n = electrons, F = Faraday constant (96485 C/mol). Combining: ΔG° = –RT ln K = –nFE°.

**Electrolysis (non-spontaneous):** Applied external voltage forces non-spontaneous redox. **Anode:** oxidation (connected to + terminal of power source). **Cathode:** reduction (connected to – terminal). **Faraday's law of electrolysis:** moles e<sup>–</sup> transferred = (Current in A × time in s) / (96485 C/mol). Use stoichiometry to find moles product or reactant.

Δ E° values are reduction potentials from tables (all half-rxns as reductions). To use oxidation: reverse the rxn AND negate E°. Multiplying coefficients doesn't change E° (it's intensive). Always use Nernst when conc ≠ 1 M.

✓ FRQ: (1) Write half-rxns (oxidation & reduction). (2) Look up E° from table. (3) E°<sub>cell</sub> = E°<sub>cath</sub> – E°<sub>anode</sub>. (4) Spont if E° > 0. (5) Nernst if non-standard conc: E = E° – (0.0592/n)logQ. (6) Link: ΔG° = –nFE°.